

Zigzag instability of columnar Taylor–Green vortices

Fr	k	σ	The growth rate from matrix-free code
1.000	2	$0.2108 + 0.0000i$	0.2117
0.500	3	$0.2251 + 0.0000i$	0.2256
0.250	5	$0.2422 + 0.0000i$	0.2423
0.125	9	$0.1998 + 0.0000i$	0.1999

Table 1. The most unstable mode at each Fr considered.

Guo J, Taylor JR, Zhou Q. Zigzag instability of columnar Taylor–Green vortices in a strongly stratified fluid. Journal of Fluid Mechanics. 2024 997:A34.

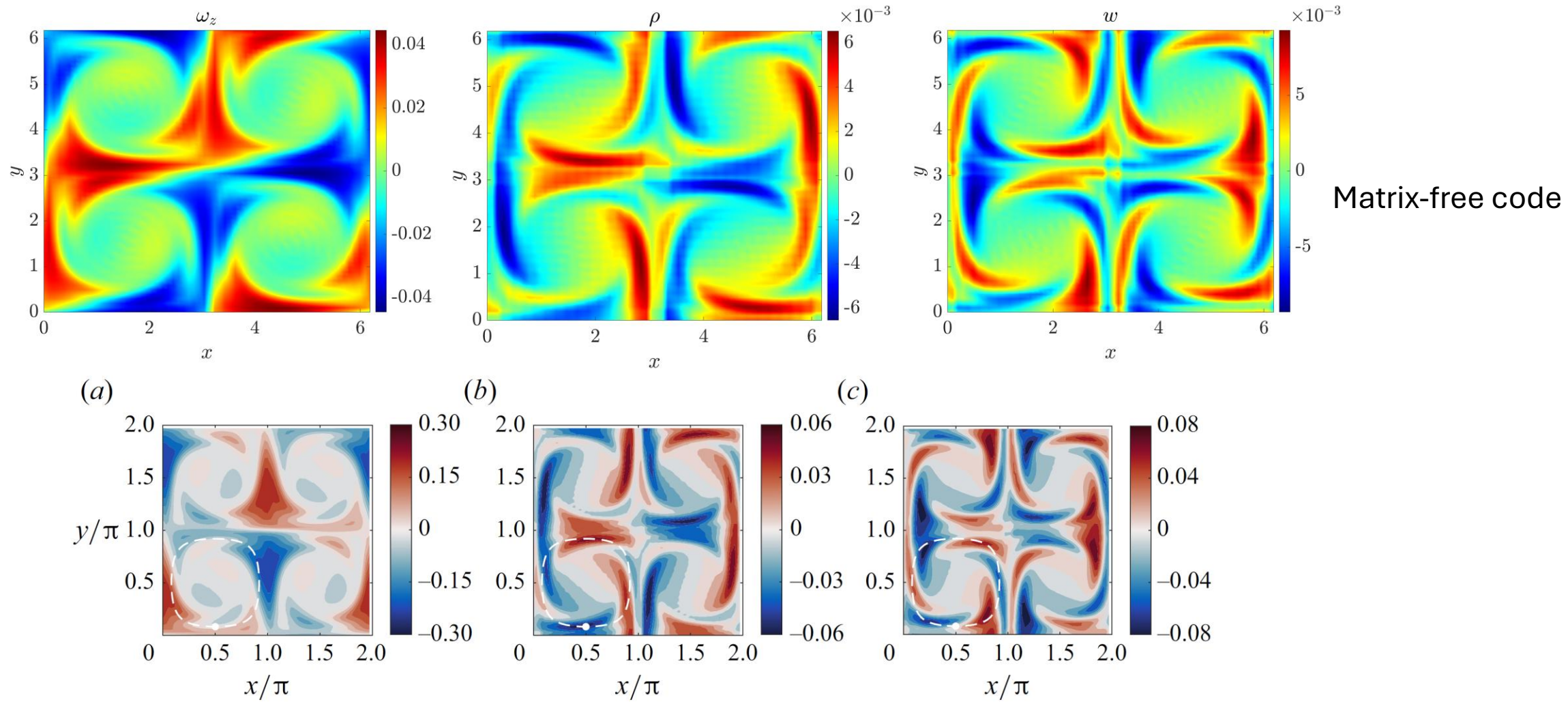
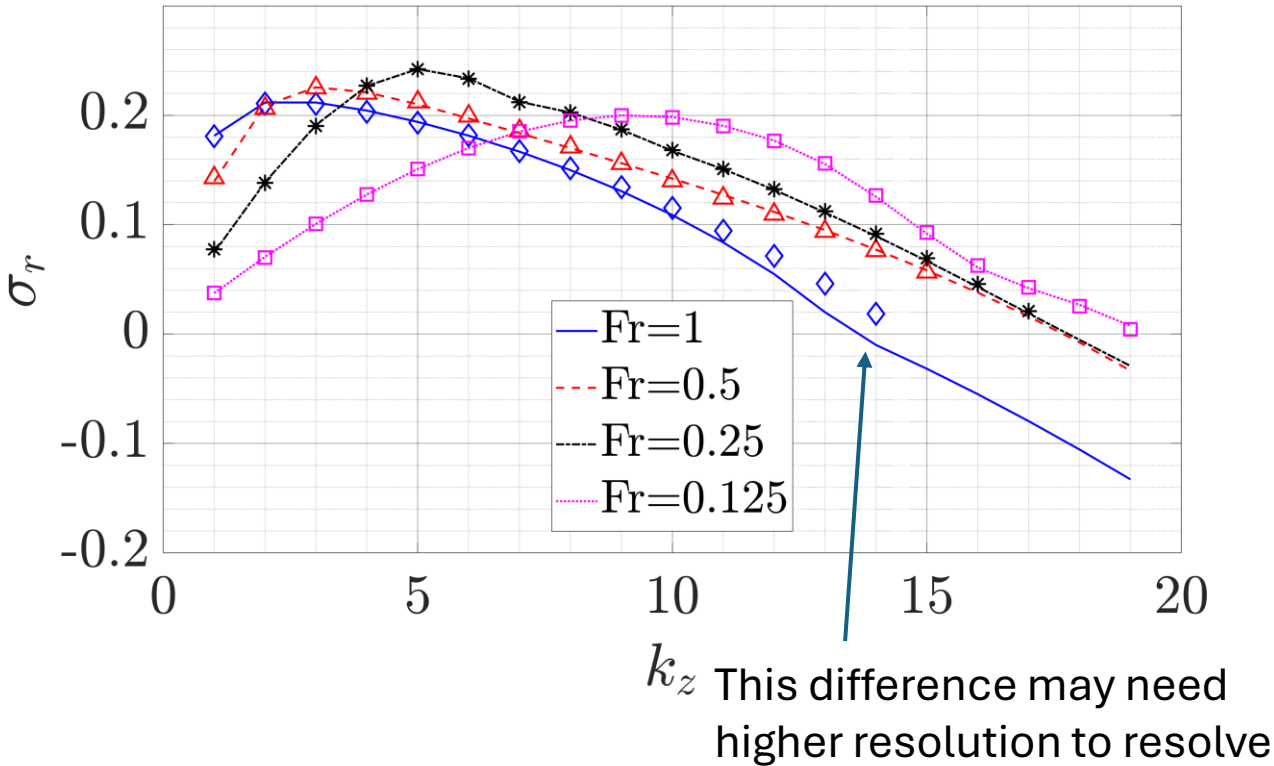
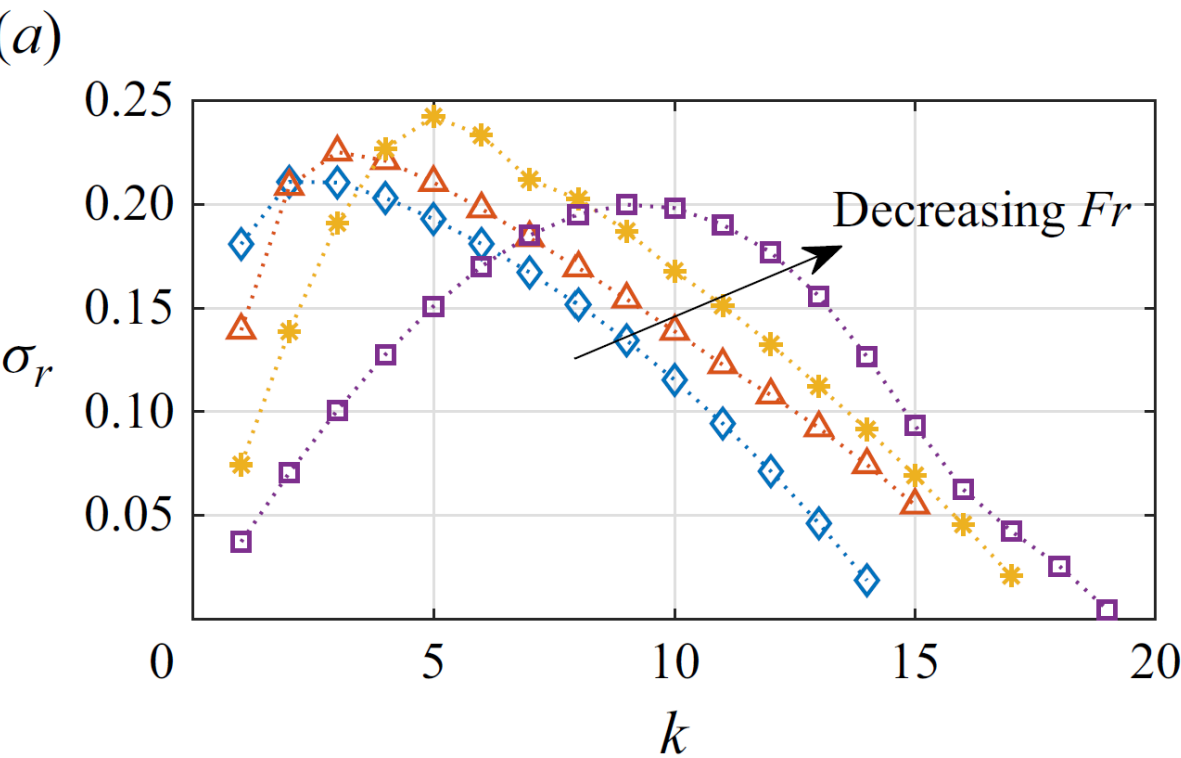


Figure 4. Horizontal transect of perturbation (a) ω_z at $z = z_0$, (b) ρ at $z = z_+$, and (c) w at $z = z_+$, for the eigenmode with $k = 3$ at $Fr = 0.5$. The heights z_0 and z_+ are specified in [figure 5](#). The amplitude of the eigenmode is normalized such that the kinetic energy associated with the horizontal perturbation velocities (u, v) constitutes 1 % of the kinetic energy in the base flow (U, V). The dashed line indicates the streamline corresponding to $\psi = 0.25$, and the solid dot indicates the starting point of a particle trajectory considered in [figure 6](#).

Figure 2(a) of Guo J, Taylor JR, Zhou Q. Zigzag instability of columnar Taylor–Green vortices in a strongly stratified fluid. Journal of Fluid Mechanics. 2024 997:A34.

Lines: matrix-free code
Markers: from Guo, Taylor & Zhou (2024)
 $N_x=N_y=64$

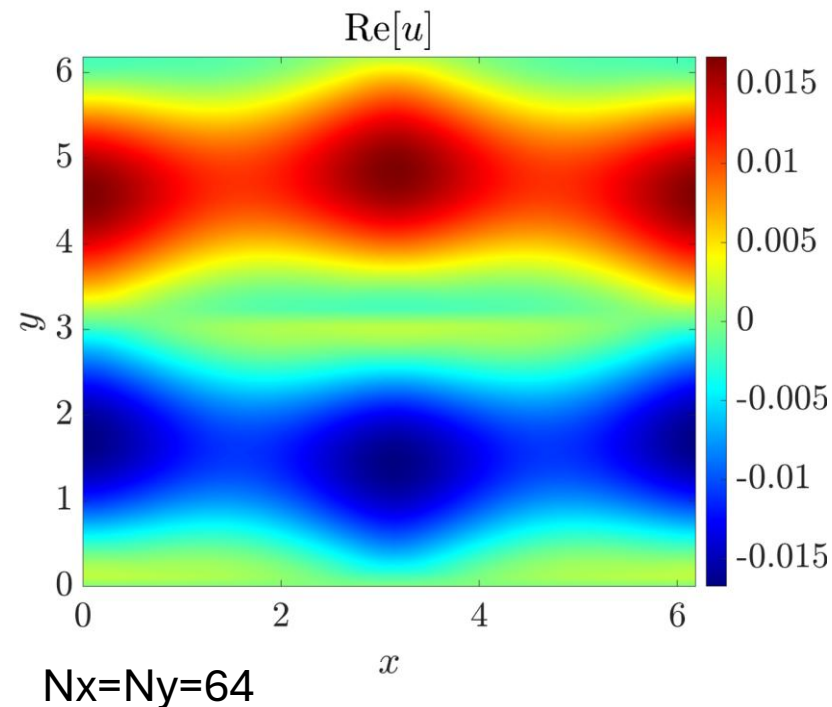
Select the stationary modes to compare growth rate



Comparing eigenvectors associated with A0_viscous, A0_adj_viscous, and A_full operators: Fr=0.125, Re=1600, Pr=0.7 (leading complex eigenmode)

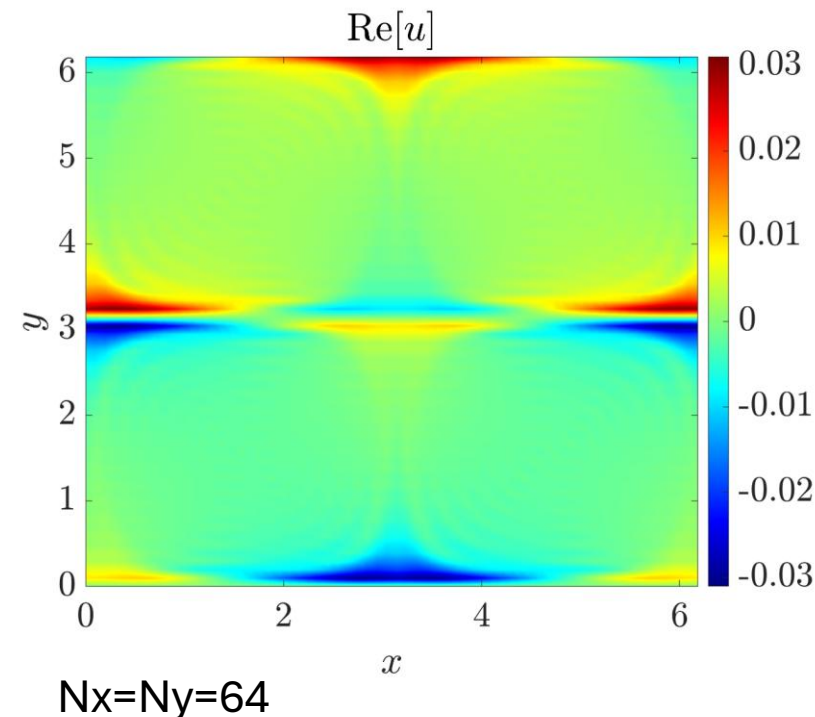
A0_viscous

lambda= 0.1409 - 0.5932i (Nx=Ny=64)



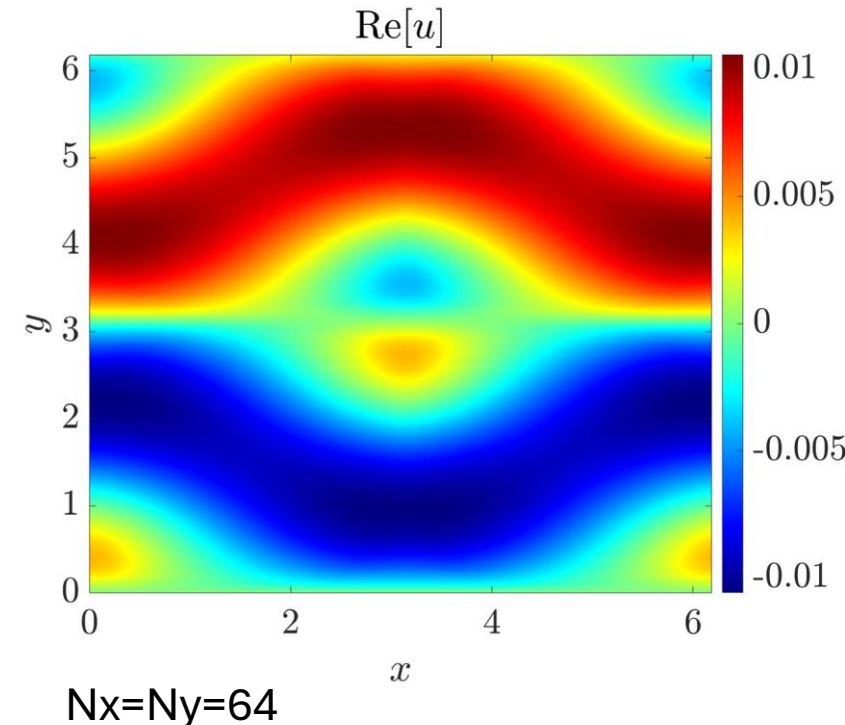
A0_adj_viscous

lambda= 0.1409 + 0.5932i (Nx=Ny=64)



A_full, with kz=0.0001

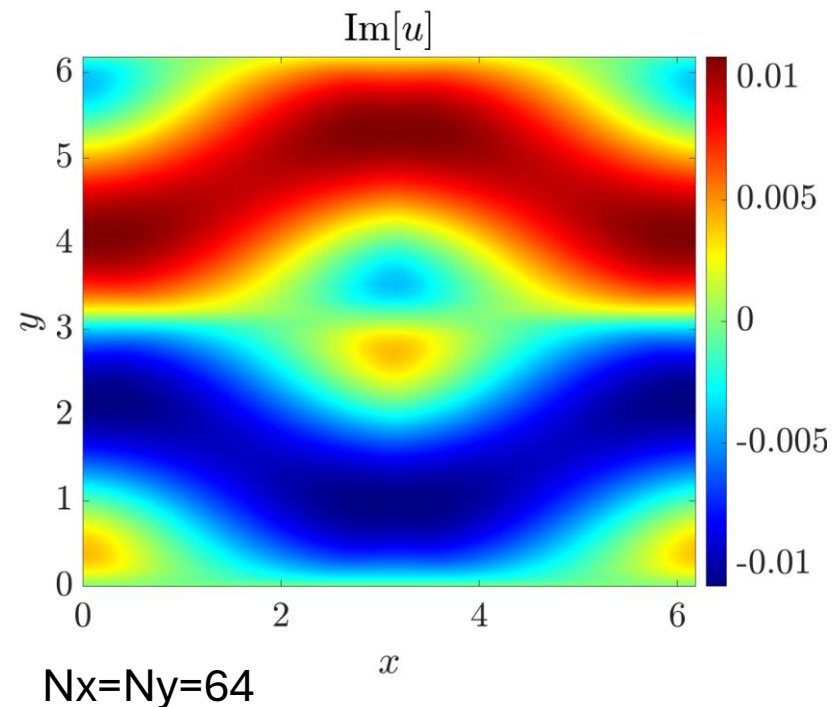
lambda= 0.1409 - 0.5932i (Nx=Ny=64)



Comparing eigenvectors associated with A0_viscous, A0_adj_viscous, and A_full operators: Fr=0.125, Re=1600, Pr=0.7 (leading complex eigenmode)

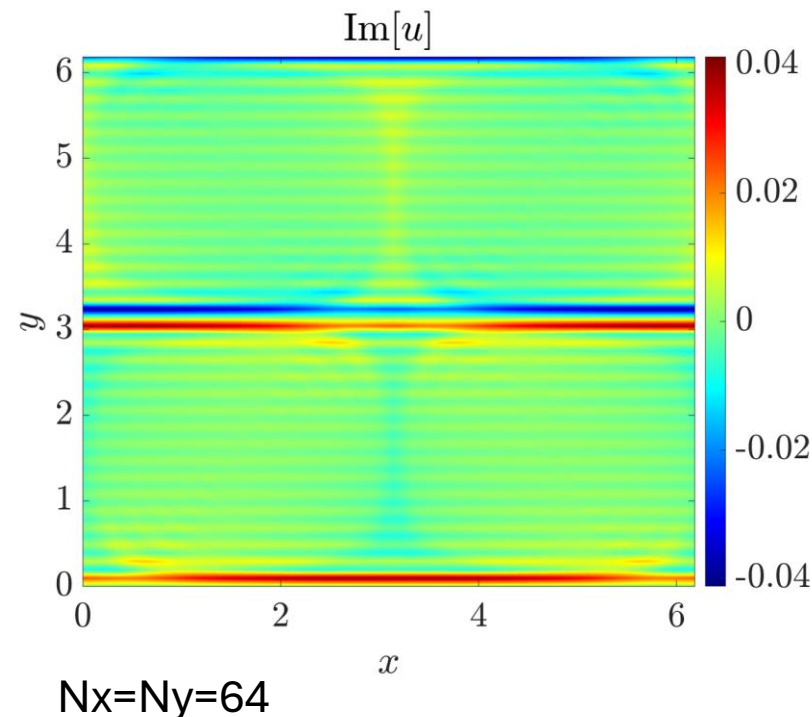
A0_viscous

lambda= 0.1409 - 0.5932i (Nx=Ny=64)



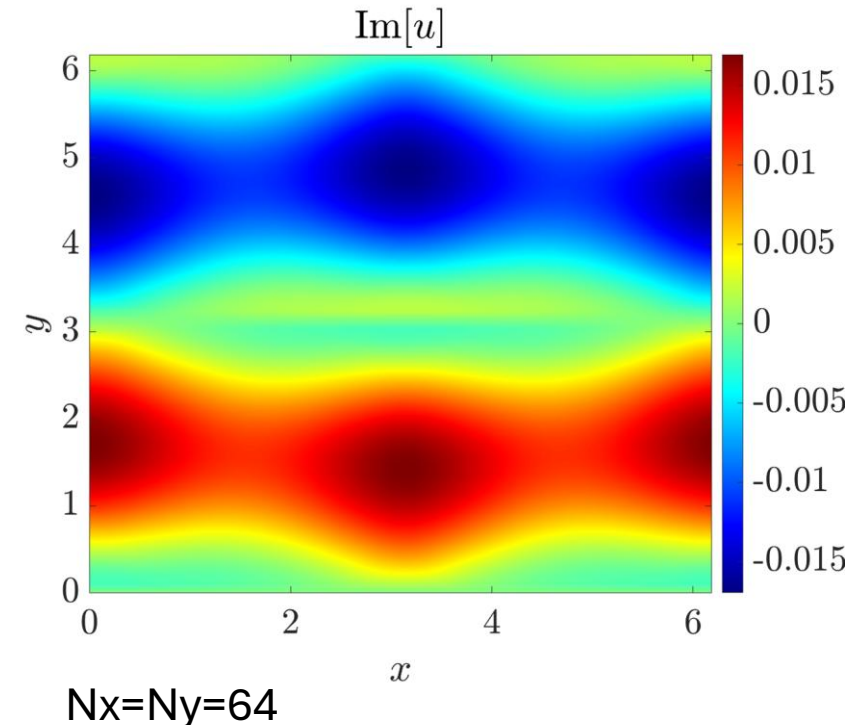
A0_adj_viscous

lambda= 0.1409 + 0.5932i (Nx=Ny=64)



A_full, with kz=0.0001

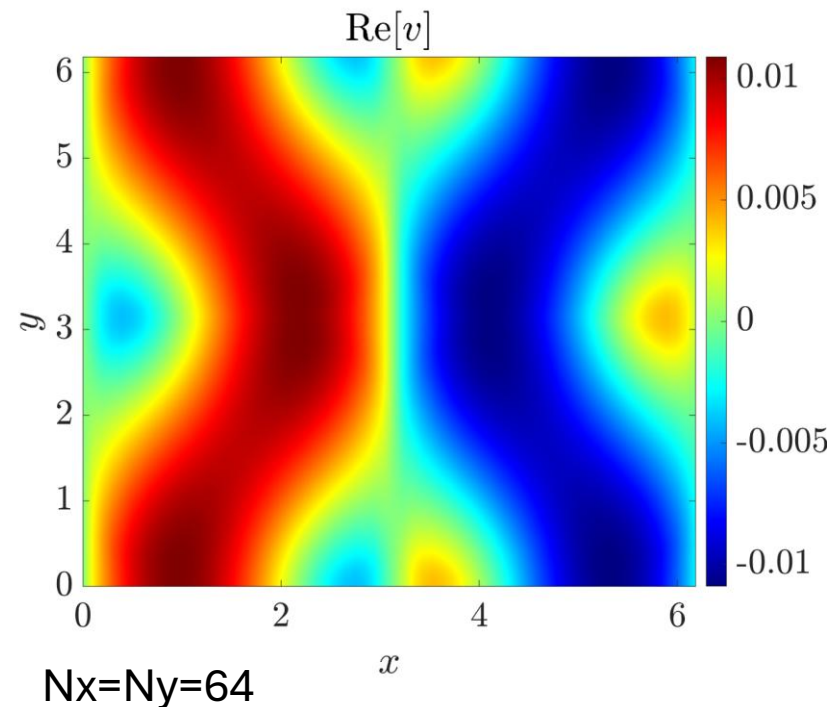
lambda= 0.1409 - 0.5932i (Nx=Ny=64)



Comparing eigenvectors associated with A0_viscous, A0_adj_viscous, and A_full operators: Fr=0.125, Re=1600, Pr=0.7 (leading complex eigenmode)

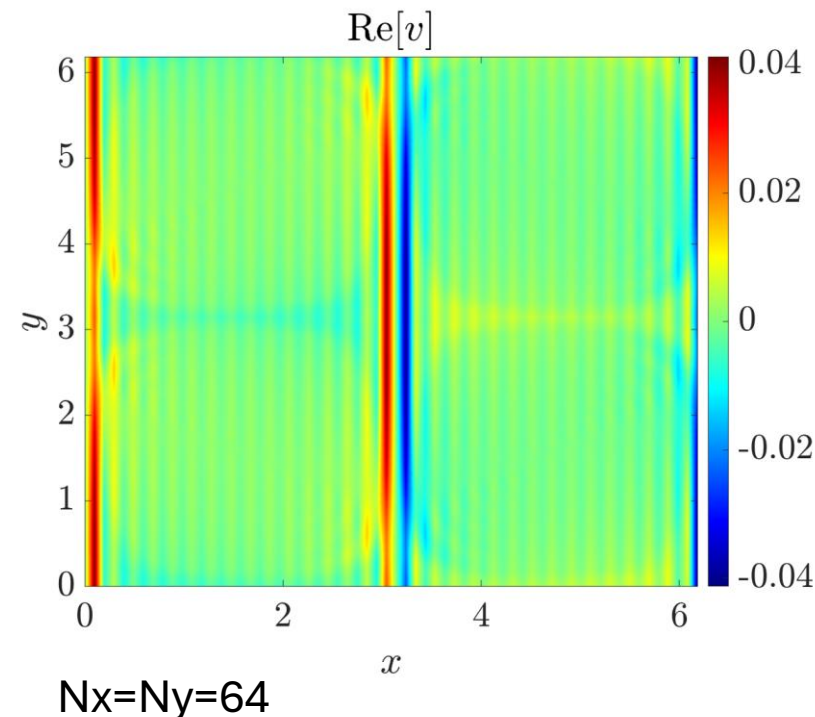
A0_viscous

lambda= 0.1409 - 0.5932i (Nx=Ny=64)



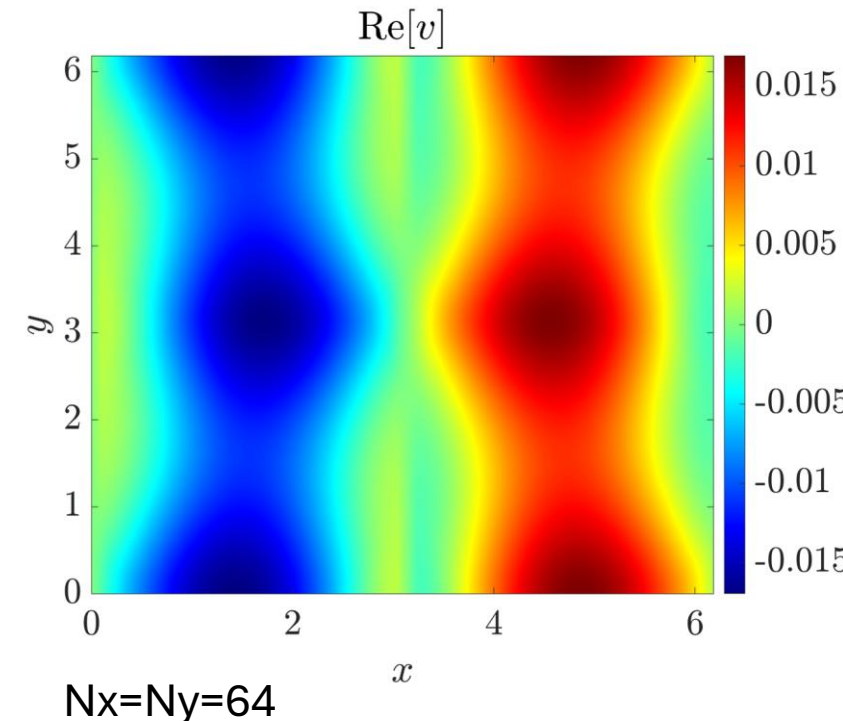
A0_adj_viscous

lambda= 0.1409 + 0.5932i (Nx=Ny=64)



A_full, with kz=0.0001

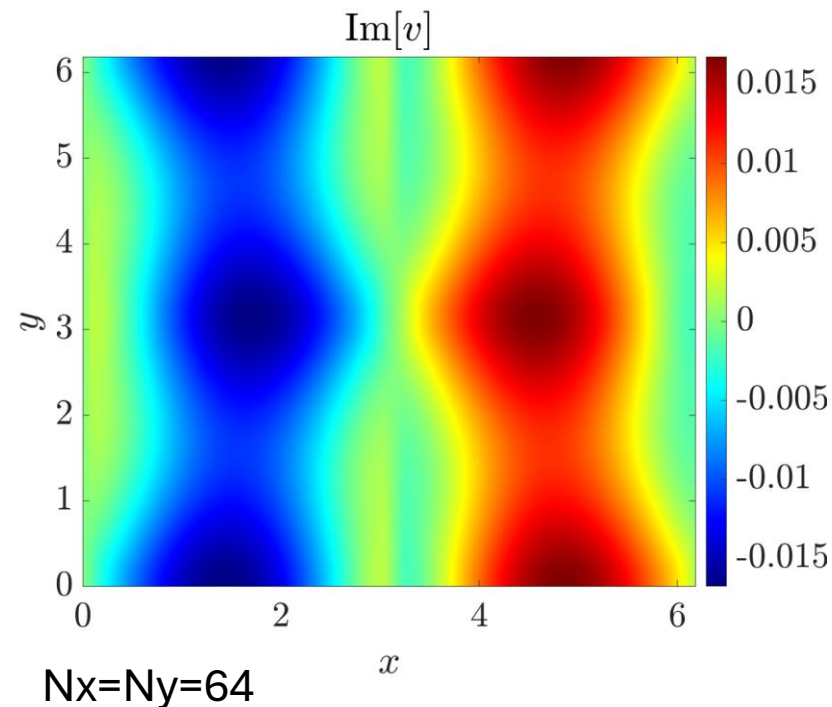
lambda= 0.1409 - 0.5932i (Nx=Ny=64)



Comparing eigenvectors associated with A0_viscous, A0_adj_viscous, and A_full operators: Fr=0.125, Re=1600, Pr=0.7 (leading complex eigenmode)

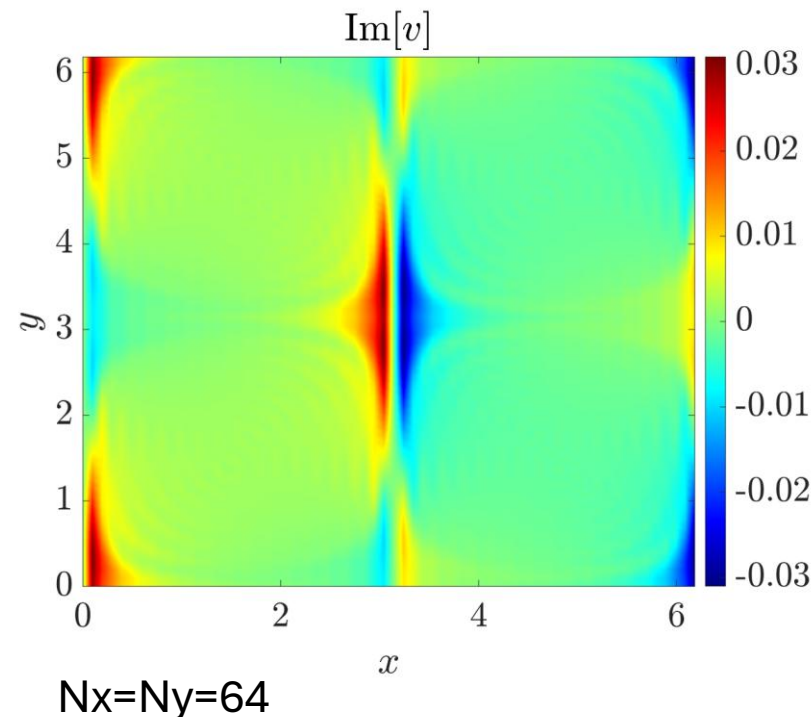
A0_viscous

lambda= 0.1409 - 0.5932i (Nx=Ny=64)



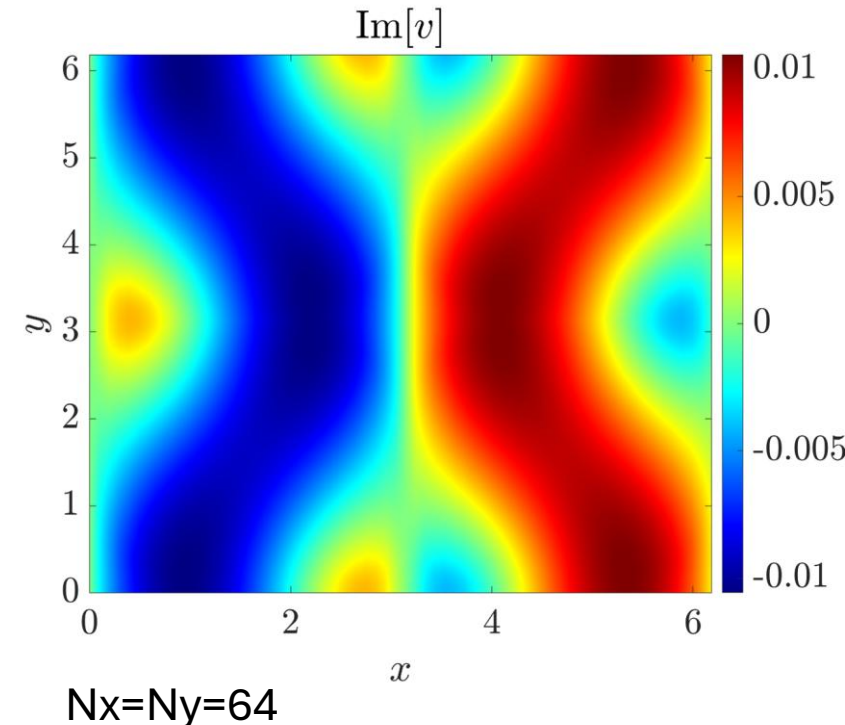
A0_adj_viscous

lambda= 0.1409 + 0.5932i (Nx=Ny=64)



A_full, with kz=0.0001

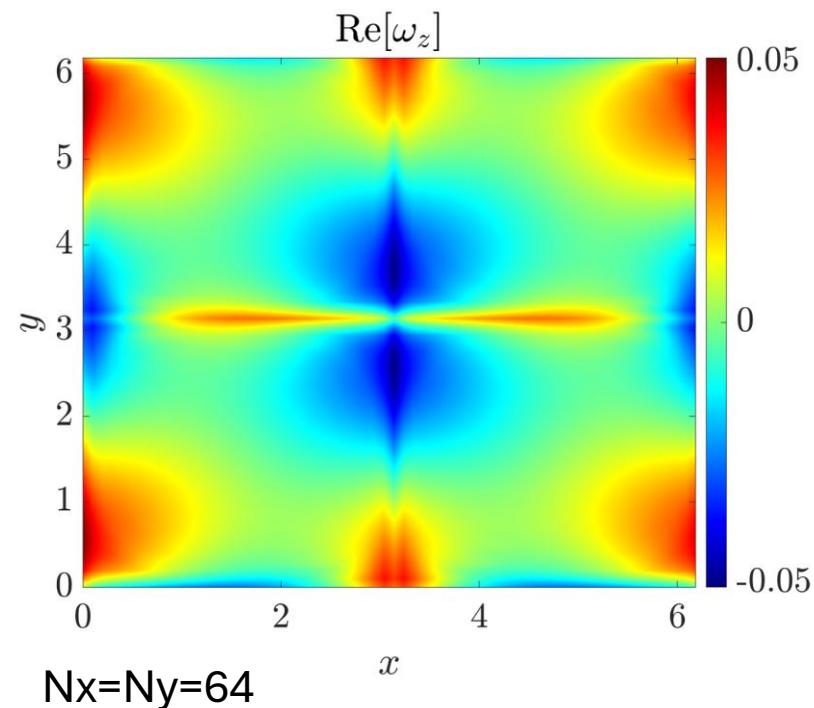
lambda= 0.1409 - 0.5932i (Nx=Ny=64)



Comparing eigenvectors associated with A0_viscous, A0_adj_viscous, and A_full operators: Fr=0.125, Re=1600, Pr=0.7 (leading complex eigenmode)

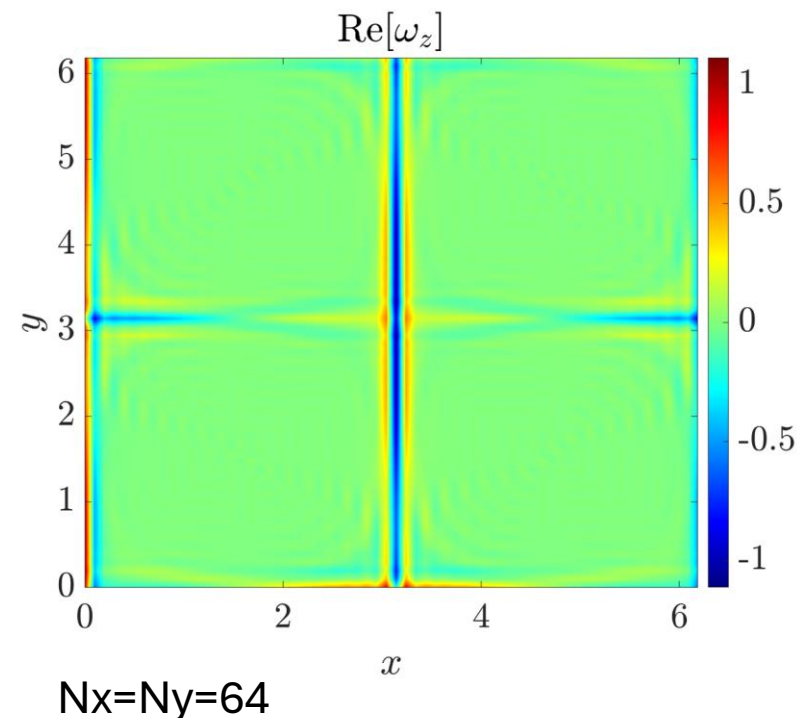
A0_viscous

lambda= 0.1409 - 0.5932i (Nx=Ny=64)



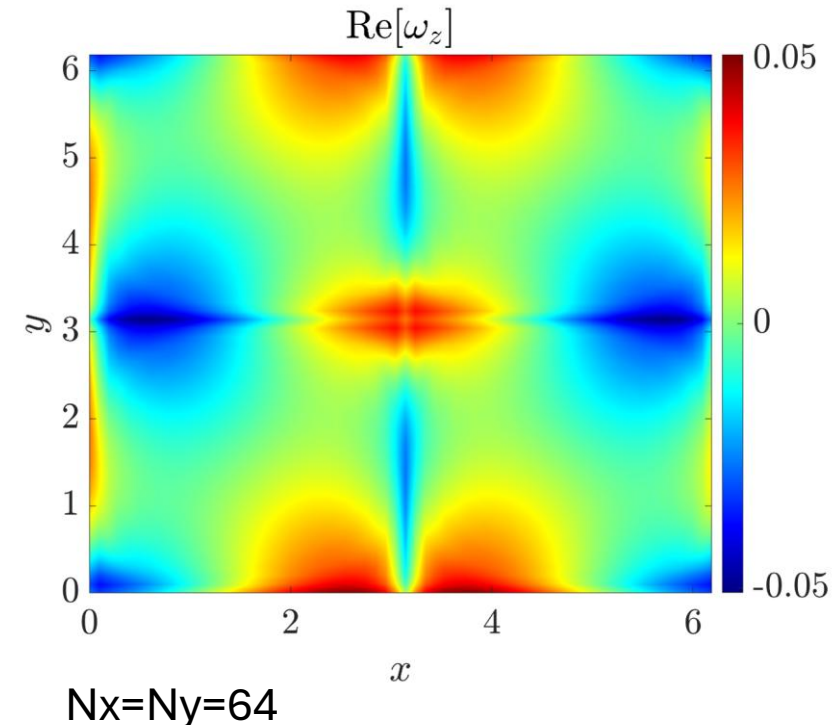
A0_adj_viscous

lambda= 0.1409 + 0.5932i (Nx=Ny=64)



A_full, with kz=0.0001

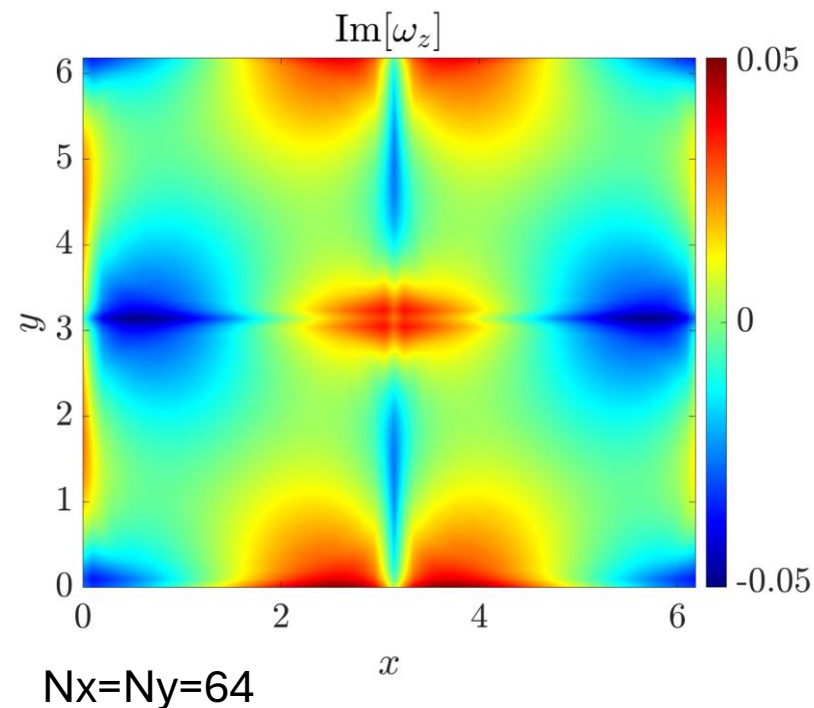
lambda= 0.1409 - 0.5932i (Nx=Ny=64)



Comparing eigenvectors associated with A0_viscous, A0_adj_viscous, and A_full operators: Fr=0.125, Re=1600, Pr=0.7 (leading complex eigenmode)

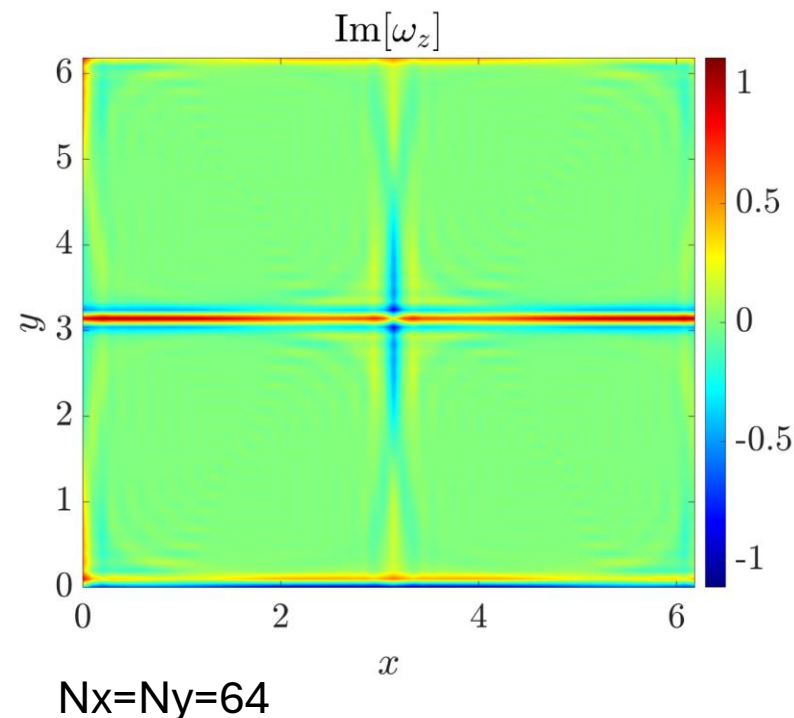
A0_viscous

lambda= 0.1409 - 0.5932i (Nx=Ny=64)



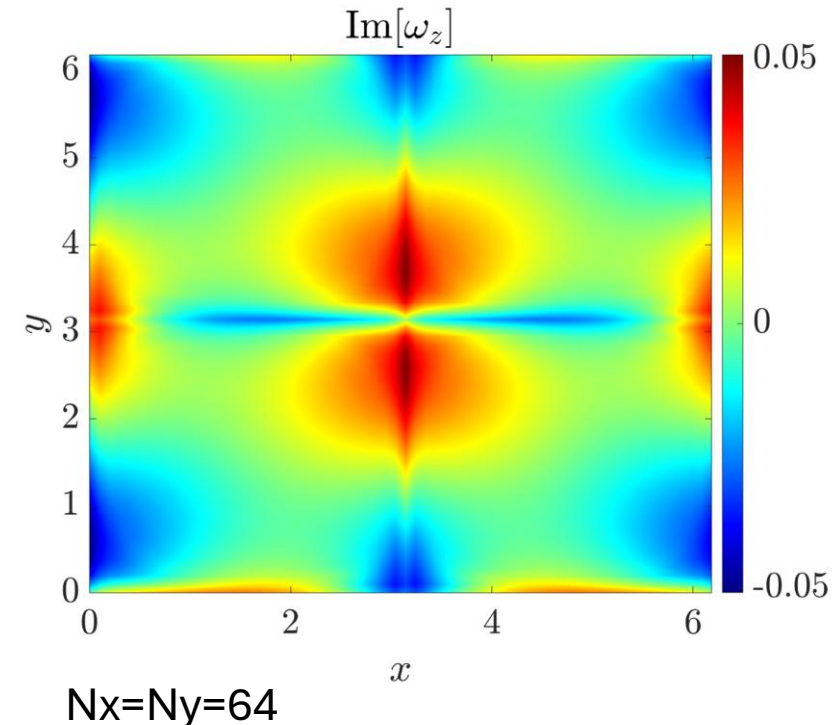
A0_adj_viscous

lambda= 0.1409 + 0.5932i (Nx=Ny=64)



A_full, with kz=0.0001

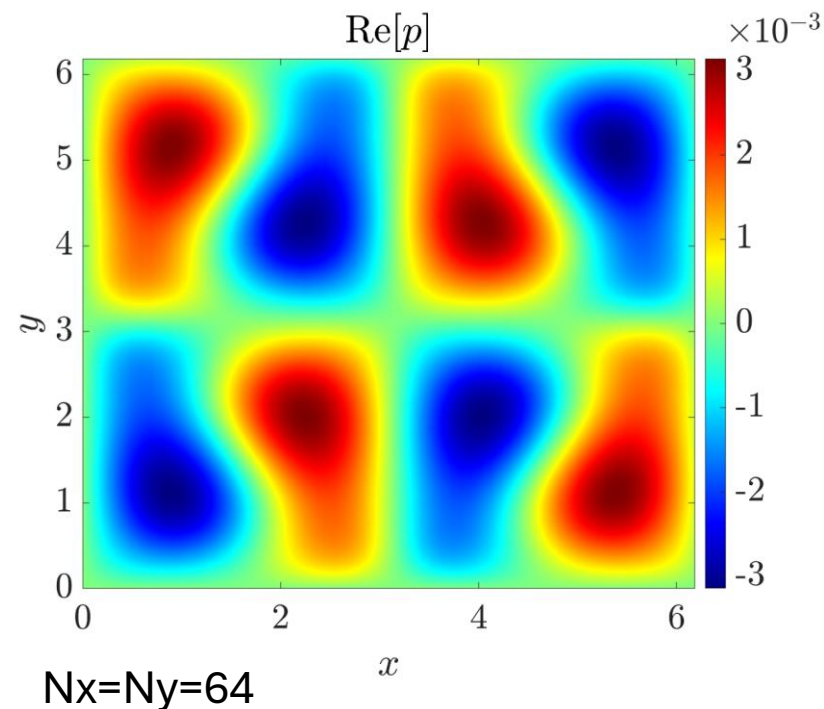
lambda= 0.1409 - 0.5932i (Nx=Ny=64)



Comparing eigenvectors associated with A0_viscous, A0_adj_viscous, and A_full operators: Fr=0.125, Re=1600, Pr=0.7 (leading complex eigenmode)

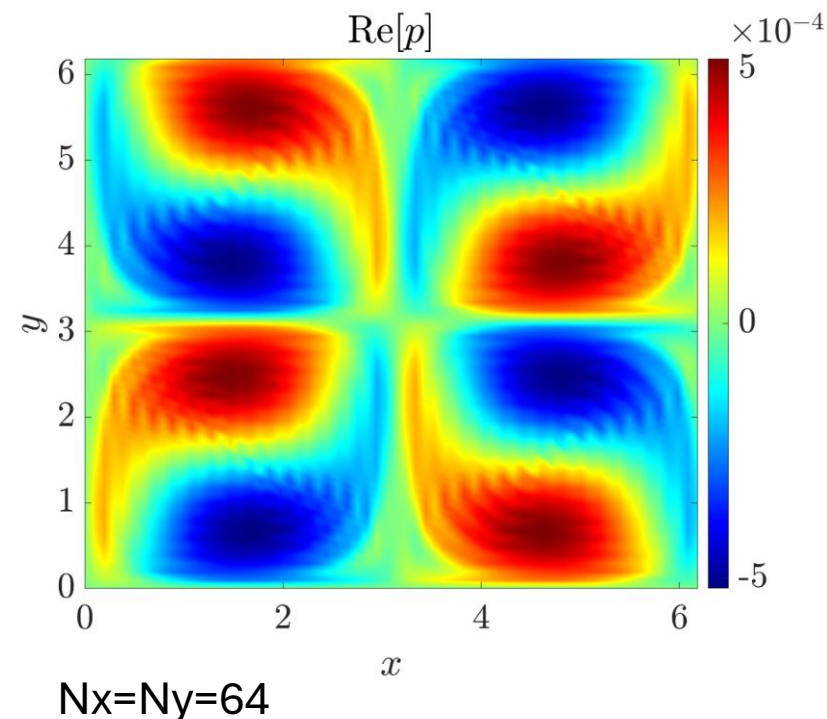
A0_viscous

lambda= 0.1409 - 0.5932i (Nx=Ny=64)



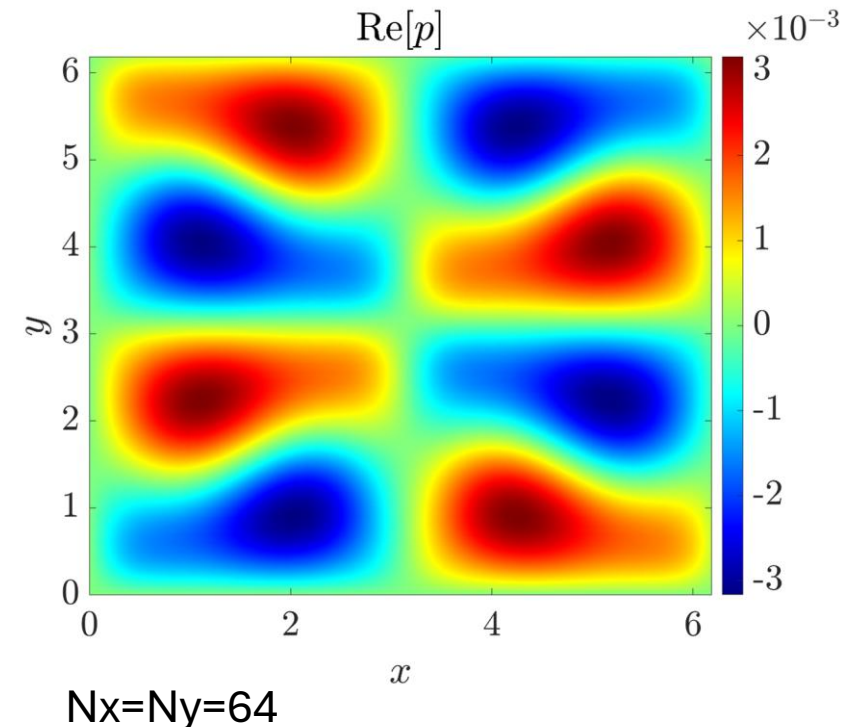
A0_adj_viscous

lambda= 0.1409 + 0.5932i (Nx=Ny=64)



A_full, with kz=0.0001

lambda= 0.1409 - 0.5932i (Nx=Ny=64)



Comparing eigenvectors associated with A0_viscous, A0_adj_viscous, and A_full operators: Fr=0.125, Re=1600, Pr=0.7 (leading complex eigenmode)

A0_viscous

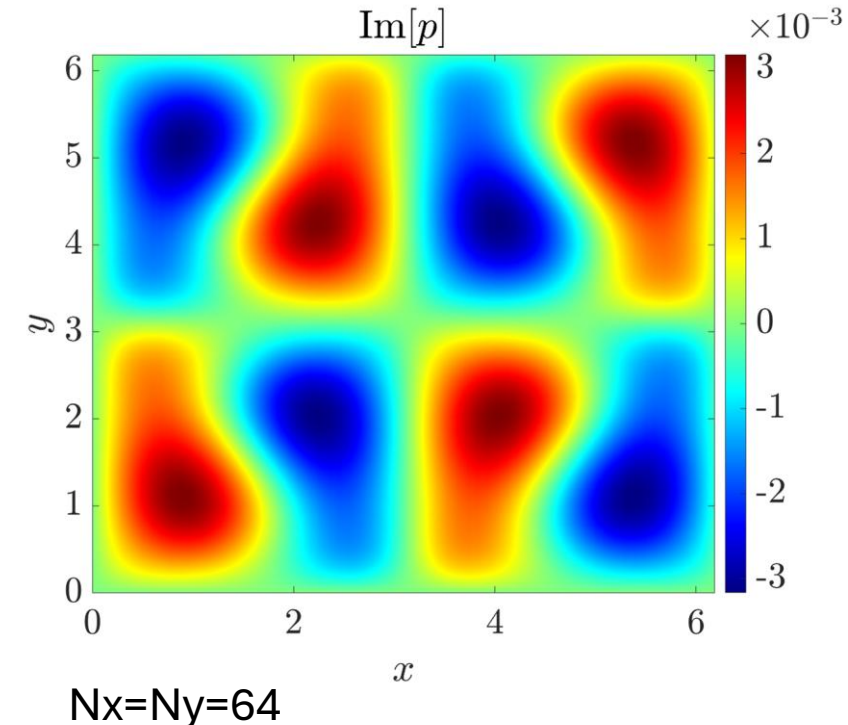
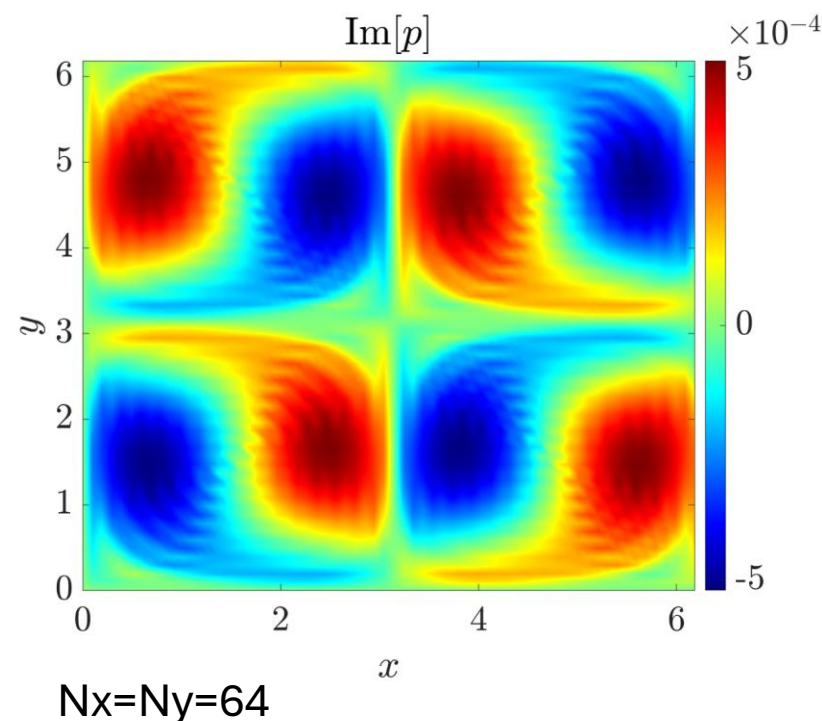
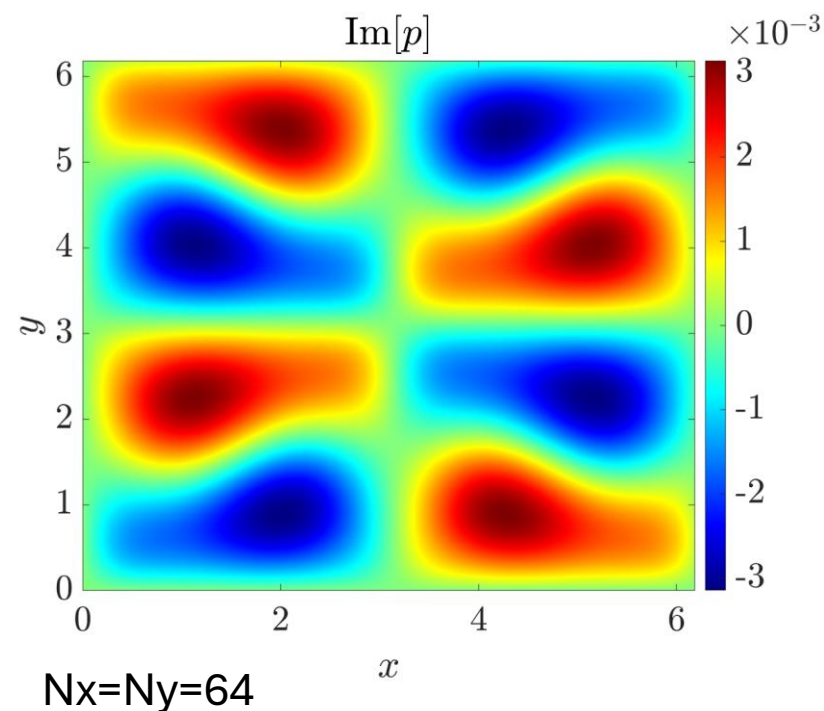
A0_adj_viscous

A_full, with kz=0.0001

lambda= 0.1409 - 0.5932i (Nx=Ny=64)

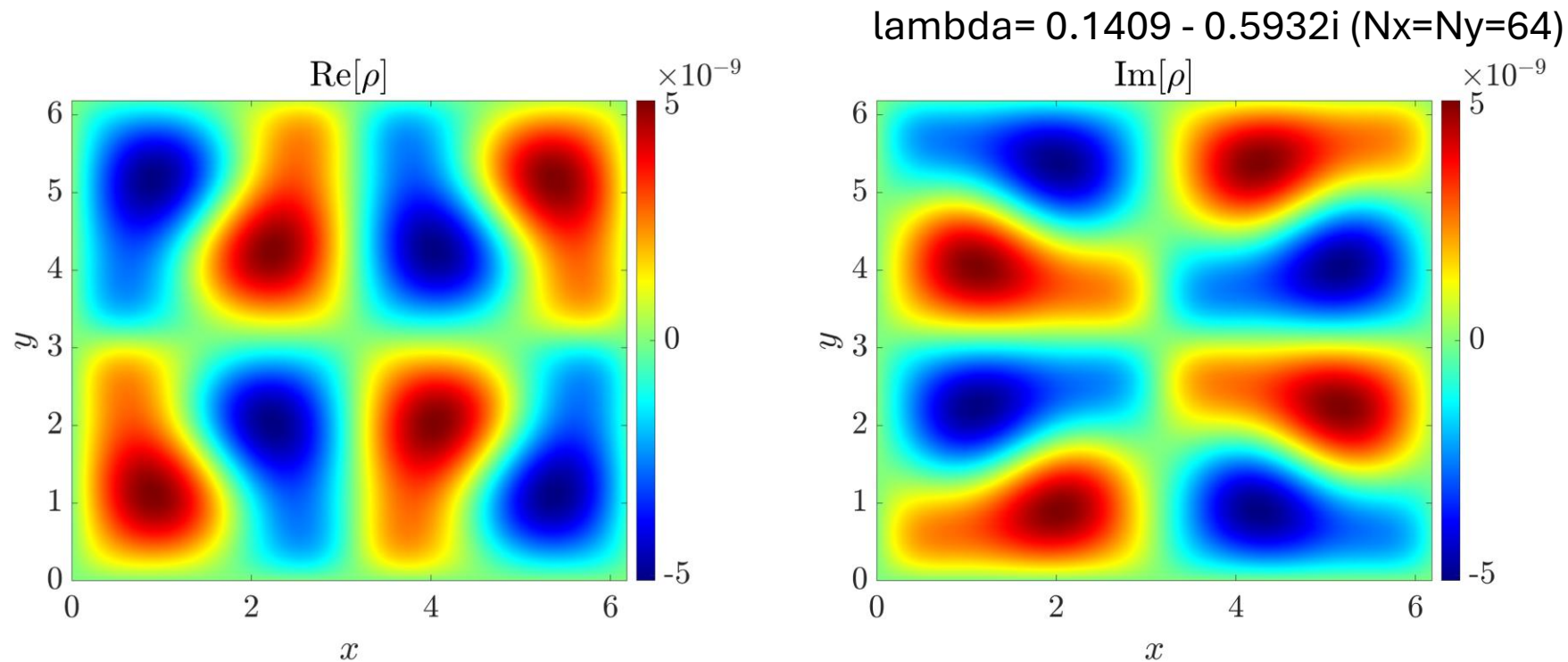
lambda= 0.1409 + 0.5932i (Nx=Ny=64)

lambda= 0.1409 - 0.5932i (Nx=Ny=64)



Comparing eigenvectors associated with $A0_viscous$, $A0_adj_viscous$, and A_full operators: $Fr=0.125$, $Re=1600$, $Pr=0.7$ (leading complex eigenmode)

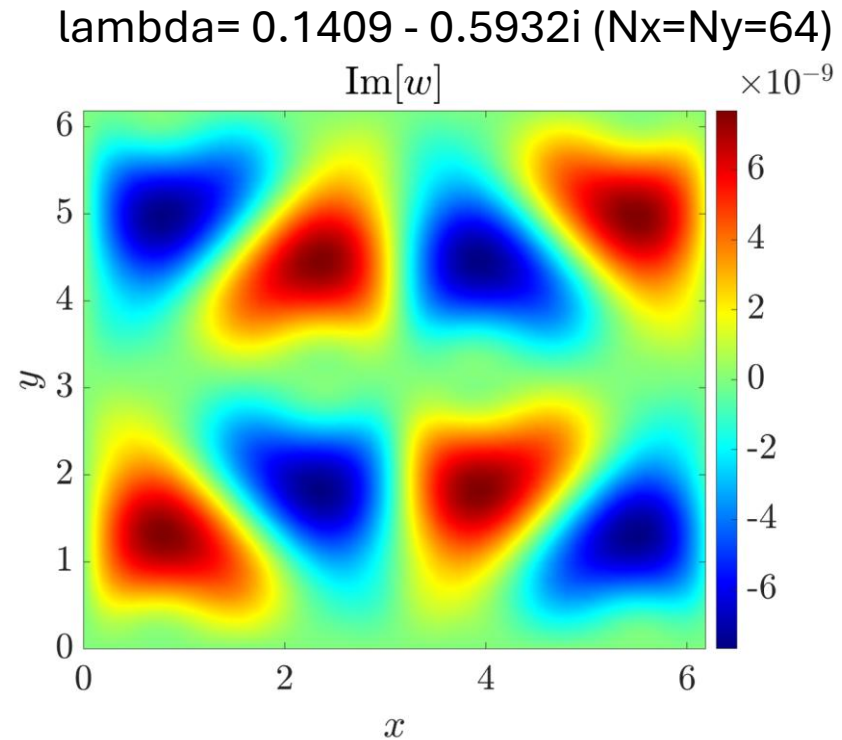
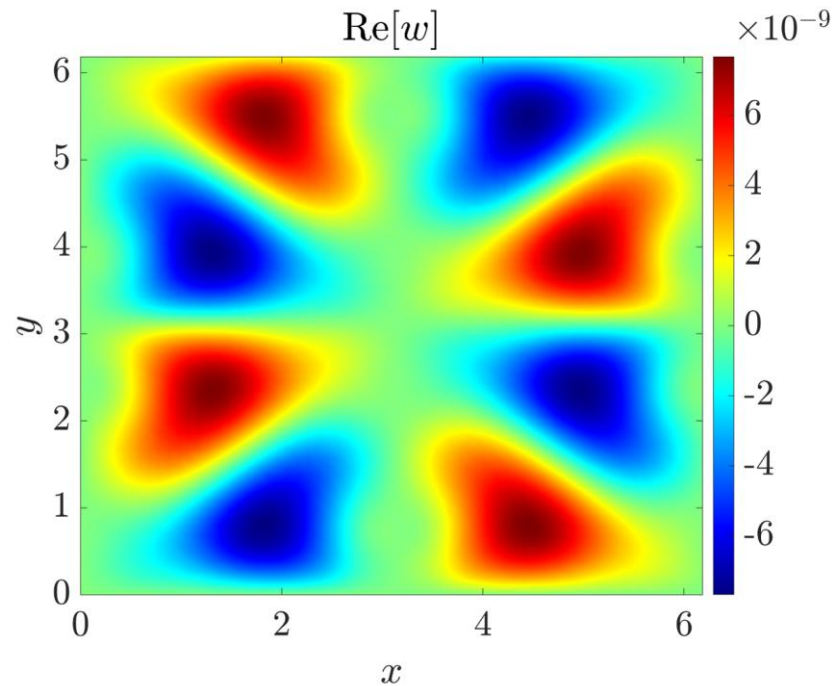
A_full , with $kz=0.0001$



$N_x=N_y=64$

Comparing eigenvectors associated with $A0_viscous$, $A0_adj_viscous$, and A_full operators: $Fr=0.125$, $Re=1600$, $Pr=0.7$ (leading complex eigenmode)

A_full , with $kz=0.0001$



$N_x=N_y=64$